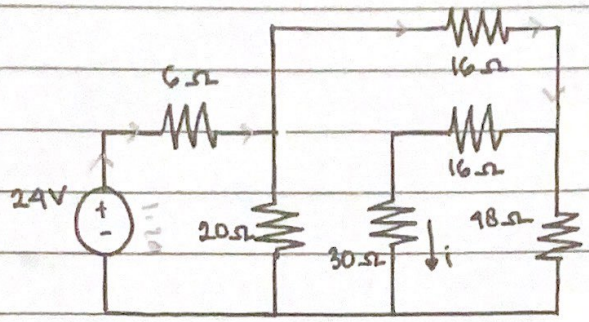
Tentukan nilai arus i !



$R_{seri_1} = 16 + 30 = 46\ \Omega$

$R_{seri_1} // 46\ \Omega \rightarrow R_{p_1} = \frac{46 \cdot 48}{46 + 48}$

$= 23,49\ \Omega$

$R_{seri_2} = 23,49 + 16 = 39,99\ \Omega$

$R_{seri_2} // 20\ \Omega \rightarrow R_{p_2} = \frac{39,99 \cdot 20}{39,99 + 20}$

$= 13,28\ \Omega$

$R_{total} = 13,28 + 6$

$= 19,28\ \Omega$

$V = I \times R$

$I = \frac{V}{R_{tot}} = \frac{24}{19,28} = 1,24\ A$

Dit : $I_{30\Omega}$?

Jawab :

$I_{30\Omega} = \frac{16 + 48}{(16 + 48) + 30} \times I_{total}$

$= \frac{64}{64 + 30} \times 1,24$

$= \frac{64}{94} \times 1,24$

$= 0,68 \times 1,24$

$= 0,8432\ A$